IP Addressing Plan

Kalahari Biodiversity Research Centre | Project ID: CMPG325-2026-107 | Client ID: CLI-107

Assigned Block

192.168.48.0/24

The assigned /24 block was split into four equal /26 subnets, one per VLAN. Equal-sized subnets were chosen for simplicity and ease of justification during the video defense, since the assigned technical challenge (DNS) does not require address-efficient VLSM subnetting.

Subnet Allocation

VLAN	Department	Subnet	Gateway	Usable Range	Broadcast
10	Admin	192.168.48.0/26	192.168.48.1	192.168.48.2 - .62	192.168.48.63
20	Research	192.168.48.64/26	192.168.48.65	192.168.48.66 - .126	192.168.48.127
30	Servers	192.168.48.128/26	192.168.48.129	192.168.48.130 - .190	192.168.48.191
40	Voice	192.168.48.192/26	192.168.48.193	192.168.48.194 - .254	192.168.48.255

Device Assignments

Device	VLAN	IP Address
Router0 - Admin sub-interface	10	192.168.48.1
Router0 - Research sub-interface	20	192.168.48.65
Router0 - Servers sub-interface	30	192.168.48.129
Router0 - Voice sub-interface	40	192.168.48.193
DNS-DHCP-SERVER	30	192.168.48.130
AdminPC1 - AdminPC3	10	192.168.48.2 - .4 (via DHCP)
ResearchPC1 - ResearchPC3	20	192.168.48.66 - .68 (via DHCP)
Admin-Phone1, Research-Phone1	40	192.168.48.194 - .195 (via DHCP)